

# Guide: Miniframes Navigation

## 1 Introduction

The `render-miniframes` function generates a navigation bar showing the progress through sections and subsections using “miniframes” (dots or squares).

Previously, it required manual structure extraction. Now, it is designed to be **configured globally** via `navigator-config`, allowing a clean one-line integration in your document.

## 2 Global Configuration

Instead of passing parameters to every `render-miniframes` call, you can set them once at the beginning of your document:

```
#import "@preview/navigator:0.1.6": navigator-config, render-miniframes

#navigator-config.update(c => {
  c.slide-selector = metadata.where(value: (t: "ContentSlide"))
  c.miniframes = (
    fill: navy,
    active-color: white,
    inactive-color: gray,
    style: "compact"
  )
  c
})

// Now you can use a simple one-liner in your header/footer:
#set page(header: context render-miniframes())
```

## 3 Function documentation

`render-miniframes(structure, current-slide-num, ...)`

### 3.1 Parameters Reference

Most parameters default to `auto`, which means they will be resolved from the global `navigator-config` state.

| Option                         | Type                                    | Effect & Expected Values                                                                                            |
|--------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>structure</code>         | <code>array</code>   <code>auto</code>  | The presentation structure. If <code>auto</code> , resolved automatically from global <code>slide-selector</code> . |
| <code>current-slide-num</code> | <code>int</code>   <code>auto</code>    | The index of the active slide. If <code>auto</code> , resolved automatically.                                       |
| <code>style</code>             | <code>string</code>   <code>auto</code> | Layout mode: <code>"compact"</code> or <code>"grid"</code> . Defaults to global config.                             |
| <code>marker-shape</code>      | <code>string</code>                     | <code>"circle"</code> (default) or <code>"square"</code> .                                                          |
| <code>marker-size</code>       | <code>length</code>                     | Diameter/width of the markers. Default: 4pt.                                                                        |
| <code>active-color</code>      | <code>color</code>   <code>auto</code>  | Color of the current slide's marker. Defaults to global config.                                                     |
| <code>inactive-color</code>    | <code>color</code>   <code>auto</code>  | Color of future slides' markers. Defaults to global config.                                                         |
| <code>fill</code>              | <code>color</code>   <code>auto</code>  | Background color of the bar block. Defaults to global config.                                                       |
| <code>text-color</code>        | <code>color</code>   <code>auto</code>  | Color of titles. Defaults to contrast with <code>fill</code> .                                                      |
| <code>text-size</code>         | <code>length</code>                     | Size of the titles. Default: 10pt.                                                                                  |

| Option             | Type               | Effect & Expected Values                                                             |
|--------------------|--------------------|--------------------------------------------------------------------------------------|
| font               | string   none      | Font family for titles. Uses document default if none.                               |
| align-mode         | string             | Global horizontal alignment of the block: "left", "center", "right".                 |
| dots-align         | string             | Alignment of the dots <b>within</b> their section column: "left", "center", "right". |
| navigation-pos     | string             | Vertical position of dots relative to titles: "top" or "bottom". Default: "bottom".  |
| show-level1-titles | bool               | Whether to display the names of sections.                                            |
| show-level2-titles | bool               | In grid mode, whether to display subsection names.                                   |
| show-numbering     | bool               | Whether to display heading numbers. Default: false.                                  |
| numbering-format   | string             | Typst numbering format string (e.g., "1.1"). Default: "1.1".                         |
| gap                | length             | Horizontal space between sections. Default: 1.5em.                                   |
| line-spacing       | length             | Vertical space between titles and dots. Default: 4pt.                                |
| inset              | dict   length      | Internal padding of the bar block. Default: (x: 1em, y: 0.5em).                      |
| radius             | length   dict      | Corner rounding of the background block. Default: 0pt.                               |
| width              | length             | Total width of the block. Default: 100%.                                             |
| outset-x           | length             | Horizontal bleed. Useful to make the bar touch page edges.                           |
| max-length         | int   dict   auto  | Maximum length before truncation. Defaults to global config.                         |
| use-short-title    | bool   dict   auto | Whether to use short titles. Defaults to global config.                              |

## 4 Structure Extraction

To work, the navigation bar needs to know the presentation structure. Two functions are provided to extract this data from metadata markers. These are called automatically by `render-miniframes()` if arguments are set to `auto`.

### 4.1 get-structure

`get-structure(slide-selector: auto, filter-selector: none)`

Scans the document for headings and slide markers. Returns a structure dictionary.

### 4.2 get-current-logical-slide-number

`get-current-logical-slide-number(slide-selector: auto, filter-selector: none)`

Determines the index of the current slide relative to the extracted structure.

### 4.3 Selection Parameters

| Option         | Type            | Description                                                                                                           |
|----------------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| slide-selector | selector   auto | The metadata type used to identify slides. Default is (t: "LogicalSlide"). Useful for custom engines (e.g., Polylux). |

| Option          | Type            | Description                                                                                                                                     |
|-----------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| filter-selector | selector   none | If provided, only pages containing this selector will be counted. Useful to exclude transition slides that might share the same slide metadata. |

## 5 Function Signature

`render-miniframes(structure, current-slide-num, ...)`

### 5.1 The structure object

The structure argument is an array of section dictionaries. Each section has the following schema:

- **Section:** (title: content, loc: location, subsections: array)
- **Subsection:** (title: content, loc: location, slides: array) OR (title: content, loc: location, subsections: array) (if 3 levels are used).
- **Slide:** (number: integer, loc: location)

#### 5.1.1 What is loc?

The loc field expects a Typst **location** object.

- **Purpose:** It defines the destination for navigation links. If a valid location is provided, clicking on the section title or the dot will take the user to that specific position in the PDF.
- **Disabling links:** If set to none, the element will be displayed normally but will not be clickable. This is used in the mock data of this guide.

### 5.2 The current-slide-num argument

An integer representing the current slide number. The function compares this value with the number field of each slide in the structure to determine its state:

- **Active:** `slide.number == current-slide-num`
- **Completed:** `slide.number < current-slide-num`
- **Future:** `slide.number > current-slide-num`

## 6 Basic usage

By default, the navigation bar uses the "grid" style and shows section titles.

```
render-miniframes(structure, 4, use-short-
title: true)
```

#### Default Grid Style

|                |                  |            |
|----------------|------------------|------------|
| 1. Intro       | 2. Methods       | 3. Results |
| 1.1 Context ●● | 2.1 Raw Data ●●● |            |
| 1.2 Goals ●    | 2.2 Tools ●      |            |

## 7 Short Titles & Truncation

Like progressive-outline, Miniframes supports short titles and truncation. This is crucial for navigation bars which have limited horizontal space.

### 7.1 Collecting Short Titles

You must pass the short titles to the structure extractor.

```
#let struct = get-structure(
  all-shorts: query(<short>)
)
```

### 7.2 Truncation & Short Titles

Then, configure the rendering. By default, use-short-title is false.

```
render-miniframes(structure, 4)
```

### 1. Original Titles (Default)

| 1. General                     | 2. Scientific                         | 3. Preliminary |
|--------------------------------|---------------------------------------|----------------|
| Introduction to<br>the Project | Methodology<br>and Data<br>Collection | Results        |
| 1.1 Background                 | 2.1 Raw Data                          |                |
| Context                        | 2.2 Analysis                          |                |
| 1.2 Objectives<br>and Goals    | Tools                                 |                |

```
render-miniframes(
  structure, 4,
  max-length: 12
)
```

### 2. Automatic Truncation

| General Intr... | Scientific M... | Preliminary ... |
|-----------------|-----------------|-----------------|
| 1.1 Backgrou... | 2.1 Raw Data    |                 |
| 1.2 Objectiv... | 2.2 Analysis... |                 |

```
render-miniframes(
  structure, 4,
  use-short-title: true
)
```

### 3. Manual Short Titles

| 1. Intro    | 2. Methods   | 3. Results |
|-------------|--------------|------------|
| 1.1 Context | 2.1 Raw Data |            |
| 1.2 Goals   | 2.2 Tools    |            |

```
render-miniframes(
  structure, 4,
  use-short-title: true,
  max-length: 8
)
```

### 4. Combined: Short Titles + Truncation

| Intro       | Methods     | Results |
|-------------|-------------|---------|
| 1.1 Cont... | 2.1 Raw ... |         |
| 1.2 Goal... | 2.2 Tool... |         |

## 8 Layout Styles

### 8.1 Compact Mode

In "compact" mode, all slide markers of a section are grouped on a single line, regardless of subsections. This is useful for saving space in the header or footer.

```
render-miniframes(
  structure, 4,
  style: 'compact',
  use-short-title: true
)
```

### Compact Mode

| 1. Intro | 2. Methods | 3. Results |
|----------|------------|------------|
|          |            |            |

### 8.2 Grid Mode

The "grid" style is ideal for presentations with many subsections, as it aligns them vertically.

```
render-miniframes(
  structure, 4,
  style: 'grid',
  show-level2-titles: true,
  use-short-title: true
)
```

### Grid Mode with Titles

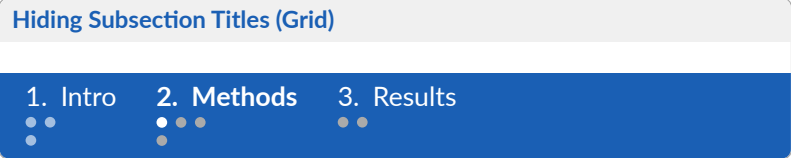
| 1. Intro    | 2. Methods   | 3. Results |
|-------------|--------------|------------|
| 1.1 Context | 2.1 Raw Data |            |
| 1.2 Goals   | 2.2 Tools    |            |

## 8.3 Hiding Titles

You can hide titles at different levels to obtain a minimalist bar.

```
render-miniframes(  
  structure, 4,  
  style: 'grid',  
  show-level2-titles: false,  
  use-short-title: true  
)
```

### Hiding Subsection Titles (Grid)

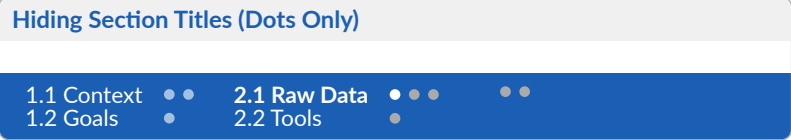


1. Intro    2. Methods    3. Results

The navigation bar has a blue background. The first level titles are '1. Intro', '2. Methods', and '3. Results'. The second level titles are hidden. The progress indicators are small dots.

```
render-miniframes(  
  structure, 4,  
  show-level1-titles: false,  
  use-short-title: true  
)
```

### Hiding Section Titles (Dots Only)



1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

The navigation bar has a blue background. The first level titles are hidden. The second level titles are '1.1 Context', '2.1 Raw Data', '1.2 Goals', and '2.2 Tools'. The progress indicators are small dots.

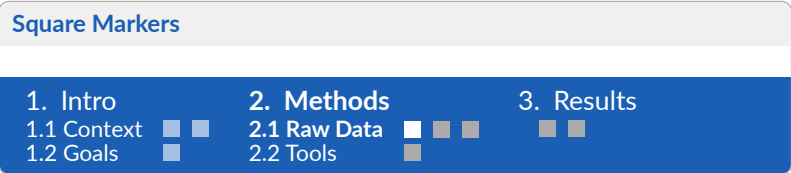
## 9 Customization

### 9.1 Markers

Change the shape and size of the progress indicators.

```
render-miniframes(  
  structure, 4,  
  marker-shape: 'square',  
  marker-size: 6pt,  
  use-short-title: true  
)
```

### Square Markers



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

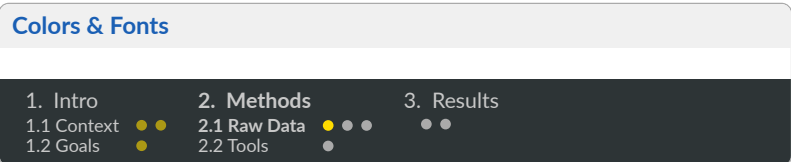
The navigation bar has a blue background. The first level titles are '1. Intro', '2. Methods', and '3. Results'. The second level titles are '1.1 Context', '2.1 Raw Data', '1.2 Goals', and '2.2 Tools'. The progress indicators are small squares.

### 9.2 Colors & Typography

Fine-tune the appearance of markers and labels.

```
render-miniframes(  
  structure, 4,  
  active-color: yellow,  
  inactive-color: gray,  
  text-color: luma(200),  
  text-size: 8pt,  
  fill: rgb('#2d3436'),  
  use-short-title: true  
)
```

### Colors & Fonts



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

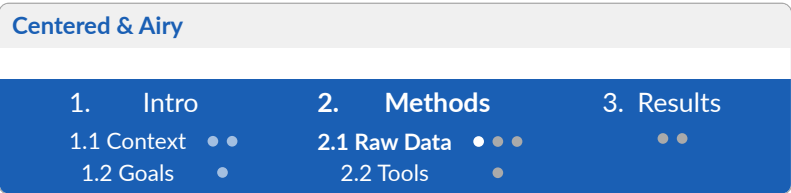
The navigation bar has a dark gray background. The first level titles are '1. Intro', '2. Methods', and '3. Results'. The second level titles are '1.1 Context', '2.1 Raw Data', '1.2 Goals', and '2.2 Tools'. The progress indicators are small circles. The text is in a light gray color.

### 9.3 Alignment & Spacing

Control the rhythm and positioning of the navigation elements.

```
render-miniframes(  
  structure, 4,  
  align-mode: 'center',  
  dots-align: 'center',  
  gap: 3em,  
  line-spacing: 8pt,  
  use-short-title: true  
)
```

### Centered & Airy



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

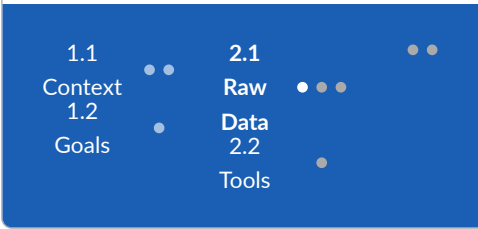
The navigation bar has a blue background. The first level titles are '1. Intro', '2. Methods', and '3. Results'. The second level titles are '1.1 Context', '2.1 Raw Data', '1.2 Goals', and '2.2 Tools'. The progress indicators are small dots. The layout is centered and airy.

### 9.4 Advanced Layout

Use inset and width to integrate the bar into specific layout zones.

```
render-miniframes(  
  structure, 4,  
  width: 60%,  
  align-mode: 'center',  
  inset: 15pt,  
  show-level1-titles: false,  
  use-short-title: true  
)
```

Compact Centered Bar



```
render-miniframes(  
  structure, 4,  
  radius: 10pt,  
  fill: rgb('#34495e'),  
  inset: (x: 2em, y: 1em),  
  use-short-title: true  
)
```

Rounded Corners

